

Active Site Predictor Proposal

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Waters

Background

Overview

The Central Dogma of Biology is the theory that information is encoded as DNA, transcribed as RNA, and translated into proteins. DNA is the basic building block of all living organisms. DNA or deoxyribose nucleic acids are a class of molecules that is comprised (for brevity) four so-called 'bases'. These bases are the molecules adenine, cytosine, guanine, and thymine, each represented by the single letter codes A, C, G, and T, respectively. DNA can then be represented by a string of these single letter codes.

Some sections of DNA encode for genes that will later be translated into 3D structures called proteins. However, to get to that point, DNA must first be transcribed into RNA or ribonucleic acid. During transcription, the transcribed DNA sequence remains relatively unchanged, but in this chemical form, the RNA molecule, a sequence that retains the same information as the DNA sequence, is able to leave the organelle where all of a cell's DNA is contained.

An RNA molecule is then able to travel into a cell's intracellular matrix, or cytoskeleton, where it then undergoes translation by a ribosome. The RNA molecule that contains the same information as the original DNA gene sequence adheres to a ribosome. The ribosome views a set of three bases along the RNA molecule and translates that set into a single amino acid. Once a set of three bases, or codon, is read and translated, the ribosome shifts three bases and translates the next codon into another amino acid. The ribosome moves along the RNA molecule in this fashion until the end of the sequence is reached. The form of the translated RNA molecule is now an amino acid chain linked by peptide bonds, this is also called a polypeptide chain.

Polypeptide chains are what proteins are made from, and each of the 20 amino acids that may be contained within any given chain are separated into their own classes based on the chemical properties they exhibit. These properties are the basis of protein folding. The polypeptide chain may start to fold into its final protein form, but will typically be transported to an additional organelle, the Golgi apparatus. In the Golgi apparatus, the polypeptide chain is correctly folded into its final form of a functional, or non-functional protein. These final proteins are the cell's workers, performing countless essential tasks your body needs to function normally, without issue. Functional proteins, or enzymes, are those that carry out some chemical process, whether that process be anabolic (creating one molecule from multiple substrate subunits) or catabolic (breaking

one substrate down into multiple).

Molecules that bind to an enzyme are called substrates, and these so-called substrates can only bind to specific regions of the enzyme's 3D structure. The regions where substrates may bind are called active sites, or catalytic centers. Substrates are able to bind to the active site of an enzyme because of the specific amino acids, or residues, placed along the surface of the protein and/or pocket of the active site. Proteins' active site are typically confirmed using technical lab techniques, but with this age of computing, predicting structures, properties, substrates, and biochemical pathways have reduced in-lab analysis.

One way to predict catalytic centers, or active sites, is to use a motif detection approach. A motif is a well-known set of neighboring residues, or amino acids, that expresses some sort of attractive force for incoming substrates. Additionally, conservation between proteins in the same protein families has become an increasingly common approach to identify and predict how a protein may be used within a biochemical pathway.

Finally, this project's purpose is to predict active sites for given proteins using Haskell as an underlying computational workhorse to predict active sites, delivering the computed data to a C++ protein viewer highlighting the top-most predicted active site candidates. Duncan Ritchie and Ian Carr will be working this semester to create a working prototype of this project detailed below.

Features

While there's really only one main feature the user can interact with, this space will be used to describe the internal process of how this tool will work.

At the command line, as the executable is called, an argument to the executable will be the accession number (a unique ID) of a protein. This accession number will be searched for and downloaded from the RCSB PDB database. The file will then be automatically stripped of all metadata and the transformation will result in only the structural data of the protein. The structural data will contain the spatial coordinates of individual atoms, the residue (molecule) those atoms are associated with, the sequence number the residue is associated with, and other unique identifiers for individual atoms.

An initial C++ (phase 1) layer will access the RCSB PDB using REST API calls paired with RCSB PDB's web API. In phase 1, the C++ script will download the respective protein file, and pass that data to a Haskell script (phase 2) that will handle all parsing. After phase 2, the parsed data will then be analyzed by other Haskell scripts (phase 3) to handle internal computations like residue mapping and motif detection. Phase 3 will also handle active site scoring, used to predict candidate active site regions. Finally, in phase 4, the computed results will be passed to another C++ script to handle protein visualization and graphics rendering using Vulkan. Vulkan is

notoriously low-level and discrete, buffers will need to be manually allocated and data will need to be manually uploaded and passed to the GPU. The resulting protein model will be rendered to the screen with a custom made low-level cross platform Vulkan forward rendered rasterization engine. The most high scoring candidate regions will then be highlighted within the protein model. The screen will be interactable, and you will be able to pan and rotate around the protein body, select which parts to render (everything, active sites only, or non-active sites only). The renderer will feature basic phong and dot based shading. Steps will be taken to ensure that this program is cross compatible with Linux and Windows devices.

Final Product

In its final form, this tool will be a command line tool that will take, as input, the accession number of the protein in question (in RCSB PDB format), and as output, display a model of the the protein with thea predicted candidate active site regions highlighted within the model.

Additionally, testing of this tool will be carried out by comparing the active sites of known, experimentally validated proteins.

Skills

Possessed

Already acquired skills that will help with the completion of this project include, but are not limited to:

- Most of the Biology theory behind the project is known by one of the group members.
- Proficiency in both C++ and Haskell languages.
- Proficiency in Haskell parsing techniques, and functional programming paradigm.
- Proficiency in graphics programming, though known libraries may be higher-level compared to Vulkan.
- Both group members are passionate about each portion of the project, (Duncan - biology theory, Ian - graphics).

To Learn

Skills necessary to learn for the completion of this project include, but are not limited to:

- Understanding build and metabuild systems to seamlessly work across different operating systems.
- Understanding Haskell and C++ FFI bridge system.
- Understanding REST APIs and the RCSB PDB web API.
- Understanding the Vulkan graphics library to visualize protein models.
- Optimizing scoring algorithm, ensuring accuracy and scientific reliability and validity.
- Understanding specific motif patterns and conservation to make reasonable scoring system.

Challenges

One major challenge will be explaining the biology theory to classmates as they may check in, ask questions, and suggest ideas or solutions. Getting the devised system to work cohesively together may prove to be a major challenge as each partner is working on different operating systems, as well as leveraging aggregated programming languages. Finally, the Vulkan ecosystem is quite low-level and will require a major time commitment to get working. Project and architecture design will be the main starting point, ensuring a modular, easy-to-debug system. Finally, Duncan has been taking time from his CS degree to focus on his minors for the past several semesters and will need to get back into the groove of programming, especially in C++.

Division of Labor

As for how the labor of the project will be broken up between members, Duncan will take on the role of Chief Algorithmic Engineer, while Ian will spearhead as the Graphics Department's Executive Director. All fun aside, a more detailed list of objectives and tasks by each team member is outlined below.

Literature Review

Similar work for the Haskell programming side includes a statistical model predicting active sites (1), that include other resources of interest, such as catalytic motifs and a basis for residue scoring (2). A separate paper includes a technique for dimensional analysis for finding clefts and pockets within or around the surface of the protein (3).

1. [Active site prediction using evolutionary and structural information](#)
2. [Analysis of Catalytic Residues in Enzyme Active Sites](#)
3. [Fpocket: An open source platform for ligand pocket detection](#)

Additionally, for the graphics section of this project, a sample section of code as a "Hello World" equivalent is available on GitHub and will be used as a reference (1). Another section of sample code showing a basic Blinn Phong shading model, while displaying the processed affect (2). Finally, a paper explains the math behind modern day perspective projection matrices (3).

1. [HelloTriangle.cc](#)
2. [Blinn Phong Model](#)
3. [Planar Geometric Projections and Viewing Transformations](#)

Stretch Goals

A few stretch goals for this project are:

- Implementing conservation scoring by fetching protein conservation data, parallel processing of amino acid sequence, and joining conservation, structural, and motif scores on individual residues.
- Implementing shadow mapping into the renderer so it is more visually appealing. Showing depth in an image is an important part of graphics programming
- Different ways to render the protein molecules. There will be the base approach which simply renders spheres, and possibly a complex one which renders helixes
- Support for Macs and Apple devices using the MoltenVK layer
- Antialiasing (MSAA or FXAA.Undecided)